

StoryBaker: Generative Memoryscapes from Personal Narratives in XR

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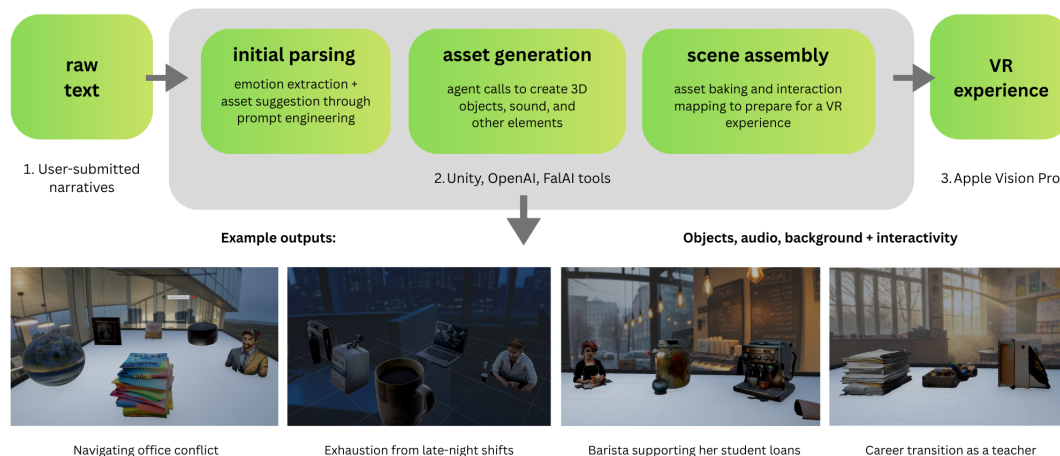


Figure 1: *StoryBaker*'s technical architecture and examples of resulting XR memoryscapes.

ABSTRACT

We introduce *StoryBaker*, a generative system for constructing memoryscapes, spatialized representations of personal narratives. *StoryBaker* enables everyday, non-technical users to externalize lived experiences as XR memoryscapes by translating narrative and emotional elements into spatial layouts, objects, sound, and interaction. Leveraging multimodal generative models and affective design frameworks, the system maps emotional and narrative cues onto spatial structures that support reflection and exploration. Through a preliminary pilot study, we probe how users engage with the generated memoryscapes of their own and others', highlighting opportunities and challenges in emotional fidelity, symbolic interaction, and perspective-taking. We discuss the implications of using XR as a medium for memory representation and outline future directions for generative approaches to spatial memory in immersive systems.

Index Terms: Memoryscapes, spatial storytelling, XR, GenAI

1 INTRODUCTION

Memoryscapes have existed in the form of VR storytelling for decades. Consumer experiences like exploring Anne Frank's objects in her virtual home [7]; collecting personal belongings and reminiscing on homelessness [1]; and witnessing a depiction of a refugee life¹ set the stage for personal reminiscing or connecting with others.

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Sharing and re-creating moments from each other's lives in VR on a personal scale, however, is rare. These experiences require prototyping, artistic, and time resources that are not available to non-technical audiences. Other storytelling media, like generative comic books², are becoming key tools for perspective-taking and empowerment through which a regular audience can share their stories and life experiences with others. However, making platforms like these more available and immersive is a challenge that the XR space has yet to address.

Existing XR tools, specifically those powered by generative artificial intelligence (GenAI) tools, enable rapid prototyping for 3D scenes for non-technical audiences. However, most generative tools focus on runtime prompts [3, 6, 11] or sandbox world creation [10, 12], rather than emotional narrative reconstruction through space and senses.

In XR, memories can be externalized as navigable spaces, allowing users to revisit, reinterpret, or share lived experiences through spatial and sensory cues. Embodiment, audio, and immersive environments let us experience virtual scenes from the first-person point of view, creating more opportunities for narrative understanding, meaningful recall, and empathy [5, 8]. Design principles of games and interactivity can also enhance spatial storytelling for virtual spaces [9]. Object symbolism, spatial metaphors, and "epistemic" actions can be used within the 3D environment and leverage the memoryscape experience.

To fill current gaps in technology and storytelling, we present *StoryBaker*, an interactive prototyping tool that democratizes access to creating and sharing personal memoryscapes. We use multiple GenAI models to aid in the creation process and allow non-technical audiences to generate audio, 3D models, and interactions related to their story.

In this work, we explore how short personal textual narratives can be transformed into spatial, affective memory representations

¹<https://www.meta.com/community/vr-for-good/the-key/>

²<https://tinyurl.com/storybook-tri>

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{
  "summary": "An office worker in their 30s feels overwhelmed by an impending career pivot.",
  "themes": ["burnout", "family", "responsibility"],
  "emotions": ["overwhelmed", "anxious", "resigned"],
  "intro_monologue": "I thought I would feel relief after making the decision, but instead everything feels heavier.",
  "sound_description": "A quiet office with distant muffled voices of a man and a woman in the background.",
  "voice_description": "Male voice, early 30s, American accent, calm but fatigued.",
  "objects": [
    "office desk",
    "chair",
    "unopened letter",
    "family portrait"
  ],
  "visual_style": "Dimly lit modern office with muted colors",
  ...
}

```

Figure 2: Example of the intermediate JSON representation to map the input narrative to spatial design elements.

in VR using generative techniques. Preliminary testing of our system reveals that participants find their memoryscapes immersive, emotional, and enhanced by spatial interactions. We see multiple applications and directions for future work, which we report on in this paper.

2 SYSTEM DESCRIPTION

We use personal memory narratives to create virtual interactive scenes in XR. To achieve this, we built a system that leverages generative resources and consists of four primary stages: (1) initial parsing, (2) agent-based asset generation, (3) scene assembly, and (4) interaction. We detail each step of the process in the following subsections.

Our technical pipelines are built using Unity 3D software for the Apple Vision Pro head-mounted display (HMD). We utilize OpenAI and FalAI APIs to parse text and generate scene assets. The end-to-end generation of a single memoryscape takes approximately 2–4 minutes. Refer to Figure 1 for a high-level overview of the system’s architecture.

2.1 Collecting memories

We accept any form of textual narrative as input to the system. Some of the examples we have demonstrated include personal stories of meaningful memories, workplace challenges, and processing difficult emotions. After the story is submitted, the system makes a series of calls that deconstruct the narrative and map it into a multisensory 3D environment.

2.2 The story parsing agent

We built a StoryParser agent that is responsible for decomposing the original raw text into spatial details. This is done through a GPT-based language model. The initial story is parsed through a model call, which we prompt-engineer based on theoretical frameworks for emotional decomposition of text. In particular, we call for Wilcox’s *Emotions Wheel* [4] and Cao et al.’s *Emotion-Aware Design* frameworks [2] in interpreting the text.

As an output from this process, we get a JSON object that includes the narrative summary, themes, suggested monologue text, voice description, ambient sound description, visual style of 3D objects, and a list of 3D objects to be included in the scene. The nodes of this JSON are later passed to corresponding generators to generate 3D assets. Figure 2 shows a simplified example of the intermediate JSON generated from parsing a memory narrative, which is used to guide multimodal asset generation.

2.3 The generator agents

The generator agents take the parsed JSON as an input and produce memoryscape assets as an output. Generators are responsible

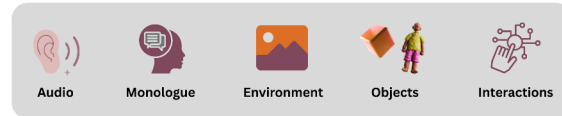


Figure 3: Narrative elements represented through XR.

for creating and populating the scene with monologue audio, background ambiance, and objects (Fig. 3). These calls include generating 2D images, 3D models, text-to-speech (TTS), and speech-to-text (STT) based on given specifications. We use multiple generative AI models for this: GPT-5-mini (text understanding), CasetteAI (sound), Minimax (TTS), Recraft (text to 2D), Trellis (2D to 3D), and Hunyuan World (360 images). These particular APIs were chosen for their creative capacity and latency.

2.4 The scene assembly agent

After the story assets are downloaded by the generator calls, they get assembled into a cohesive XR memoryscape. Audio files are attached to the necessary objects, and interaction scripts are added. For this step, we created a reusable script blueprint that “bakes” the downloaded assets into a single scene, linking helper scripts onto the scene objects. This is responsible for adding (1) default XR interactables to 3D objects; (2) special interactions for additional emotional or spatial effects, such as shattering for a “rage” room-like experience; and (3) the dialogue agent, which takes the context of the story and the scene, and can answer questions spoken out loud during the memoryscape view in real-time. This is true for any given scene generated with the system.

2.5 The XR experience

Any given scene starts with a monologue introduction of the memoryscape. Users can walk around to view the generated world around them, grab objects, and have spoken interactions with the dialogue agent to follow up about any detail of the world. The generated objects in the world serve as storytelling vehicles, revealing additional relevant details through monologues played on interaction. Other baked-in XR interactions serve as additional metaphorical and symbolic elements for different emotions in the story; for example, when the narrative is about something particularly fragile or is led with anger.

Users can interact with the dialogue agent by verbally asking a question when viewing a scene. It generates responses in real-time with the voice that the rest of the story is told with. The agent is trained to answer based on the original text of the story and has context around how the scene was generated, such as which emotions are dominating, and which assets are generated and why. Example outputs of our system are shown in Figure 1.

3 DESIGN IMPLICATIONS FROM EARLY TESTING

We have piloted our system with a group of three early participants who (1) shared a personal text-story and viewed it as a VR scene and (2) viewed a second VR scene created by another participant. Our protocol was approved by an Institutional Review Board (IRB) external to our organization. Additionally, three other participants have shared their stories and generated scenes using our system, but have not participated in an in-depth interview.

We targeted the Apple Vision Pro HMD as our XR device due to its high-resolution display, interaction toolkit, and support for rapid deployment during development. Early testing of *StoryBaker* provides insight into the opportunities and challenges for generative XR memoryscapes, which we detail below.



Figure 4: System demonstration with pilot participants.

3.1 Memories and emotions

Emotional accuracy played an important role in XR memoryscape representations. Participants' responses indicate that memoryscapes generated by our system do capture key emotional cues of a narrative. However, occasional mismatches between remembered emotion and spatial metaphor did happen and produced confusion. P1 shared a narrative about participating in and winning a hackathon, which included both stress and joy. Our system focused mainly on the stress emotions and less on the positive emotions around winning the hackathon, which was an important part of the original experience. P1 said: *"I think [the scene] was kind of aligned ... [but] the objects in my scene were a little bit different from what actually [happened]...I think the stress of doing the hackathon was definitely [present]."*

Emotional alignment, therefore, plays a critical role in how users recognize and relate to a spatialized memory of the self and others. When emotional accuracy diverges from the author's lived experience, the resulting memoryscape may be interpreted as an interpretation of an event, rather than a faithful recollection. This can be beneficial in artistic storytelling, where the focus on emotion is key. However, if the emphasis is on the factual accuracy of a memory, XR systems will need more author participation in the design process and personalization beyond theoretical frameworks. Both emotional alignment and author-centric tools can be reflected through specifications of audio, light, and other assets for such systems.

3.2 Narration and dialogue

There were multiple narration elements throughout the XR experience: the introduction monologue, the object clues, and the dialogue responses. P2 mentions the benefit of this experience in setting the tone to somebody else's memoryscape and the ability to relate: *"I think when the narrator starts to tell the stories I can relate to that. I feel like there's certain emotional aspects in the way [the] narrator talks about the stories."* Some participants expressed trust implications over the narrator's overly cinematic voice, P1: *"I think his voice was a little bit too movie[-]like."*

Additionally, participants engaged with the dialogue agent as a proxy for the original storyteller, using conversation to clarify context, motivations, and emotional nuances that were not immediately apparent from spatial cues alone. P2 appreciated the accuracy when interacting with the narrator: *"I think the agent...repeated my story. So, I would say yeah...so far it seems to be trustworthy."*

Vocal narration and dialogue can support memory interpretation in XR by supplementing spatial representations with reflective and explanatory context. Future work should explore how such systems can accurately represent the author's voice or serve as trusted proxy agents during the interaction process. Conversation mode may additionally be helpful if a memoryscape serves as a recall exercise or an educational experience.

3.3 Meaningful actions

Finally, interacting with the 3D space in a meaningful way is one of the key design advantages of XR. Participants consistently sought to probe the space via object manipulation, dialogue with the narrator agent, or exploratory behaviors to clarify or reinterpret narrative elements. To P1, interactions were a part of a joyful learning experience, *"It was fun picking up and dragging stuff down. It was really funny when things fell off the map..."* In addition to default XR interactions, P1 probed to extrapolate non-trivial interactions, such as connecting objects in the scene, to see if that would trigger any new information. Other participants also expressed interest in puzzle-like, escape-room memoryscapes that would let them re-experience certain memories and use embodied interactions for emotional release, such as breaking virtual objects.

Beyond entertainment and novelty, this points to a need for richer symbolic interactions and potentially structured interaction grammars (e.g., puzzle mechanics, investigative actions) that align with the memory and emotion of the story. These interactions can function as additional mechanisms for memory exploration and spatial metaphor through symbolic actions and interactions.

4 FINAL REFLECTIONS AND CONCLUSION

It is important to note that we target short-form personal textual narratives that are unlikely to receive the time, expertise, or funding necessary for hand-crafted XR production. These include everyday challenges, reflections, or lived experiences that, nonetheless, can and should be shared with others. *StoryBaker* makes this possible.

There are several design opportunities that we foresee for our system. Re-creating memories is a nuanced and complex process in XR; representing avatars, voice, and emotion creates questions of how realistic certain virtual assets should be, and whether generative AI can do justice in portraying the story and narrator.

Prototyping systems that are ethical, human-centered, and author-accurate are key for generative memoryscapes. Authors should be given opportunity to modify the story as they see fit and have more agency over the world design process. These systems should provide more transparency around prompts and generative options to customize the output. Additional applications of such systems include oral histories, fictional vignettes, or longer, interconnected narratives that broaden the applicability of storytelling and memory XR tools.

In this work, we introduced our vision and technical approach to generative memoryscape creation in XR from short personal narratives. We use a collection of custom AI agents to parse the original textual narrative, generate audio and 3D assets, and assemble these assets into a cohesive interactive scene on an XR HMD. Through a preliminary pilot, we explored how users engage with spatialized representations of their own and others' memories. Our early findings suggest that spatial memory in XR is inherently interpretive and is shaped by emotional cues, symbolic interactions, and narrative context.

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